

Rensselaer Polytechnic Institute
Department of Electrical, Computer, and Systems Engineering
ECSE 2500: Engineering Probability, Spring 2026

Homework #7: due Thursday, April 9th, at 2PM.

Show all work for full credit!

Submit your work as a single PDF on Gradescope, labeling each problem number with a page.
While you can check your answers with Wolfram Alpha, remember that you're going to have to do integrals by hand on the exam!

1. (30 points.) From looking at a player's gaming behavior, we try to classify whether they are playing **Hades 2 (H)** or **Silksong (S)**. By looking at Steam statistics, we estimate the prior probabilities:

$$P(H) = 0.6, \quad P(S) = 0.4.$$

We extract the following features from the players' behavior (assume all features are conditionally independent):

- X = number of player deaths per hour, modeled as a class-conditional Gaussian random variable with parameters:

$$(\mu_H = 5.0, \sigma_H^2 = 1.0) \quad \text{if playing Hades,} \quad (\mu_S = 4.5, \sigma_S^2 = 1.2) \quad \text{if playing Silksong.}$$

- Y = total hours played so far, modeled as a class-conditional exponential random variable with parameter:

$$\lambda_H = 0.25 \quad \text{for Hades,} \quad \lambda_S = 0.35 \quad \text{for Silksong.}$$

- Z = indicator of whether the player has broken their controller in frustration (1 = yes, 0 = no):

$$P(Z = 1 | H) = 0.15, \quad P(Z = 1 | S) = 0.25.$$

We now observe that a particular player has died an average of 5.1 times per hour, played 3.2 hours so far, and has broken their controller in frustration.

- (a) (20 points) What game do we decide the gamer is playing if we apply the Maximum Likelihood Decision Rule?
- (b) (10 points) What game do we decide the gamer is playing if we apply the the Bayes Decision Rule?

2. (30 points) Let X and Y be defined by the joint PDF below (note this is the same PDF we used in the last problem of Homework 6):

$$f_{XY}(x, y) = \begin{cases} \frac{6}{19}(x^2 + y^3) & 0 \leq x \leq 2, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- (a) (10 points) Compute $f_{X|Y}(x|y)$. (You need to compute the marginal $f_Y(y)$ along the way.)
- (b) (10 points) Compute $E(X|Y=y)$.
- (c) (10 points) Compute $E(X)$ using the law of iterated expectations.

More fun on the next page →

3. (20 points) Stilgar selects a sample X from an exponential distribution with $\lambda = 3$. Chani then selects a sample Y from an exponential distribution with $\lambda = X$. Finally, they ingest Y pounds of spice. Compute the PDF of Y . (Note: you may need an integration table, integration by parts, or WolframAlpha.)
4. (20 points) Anthropic is considering a subscription service in which its customers pay a flat rate for as many Claude code generation requests as they want during a given week. Suppose Anthropic's internal cost for code generation i (GPU/power/network usage, etc.) is X_i , a uniform random variable on $[1, 4]$ dollars. Let Y_i be the (continuous) number of minutes between a customer's request $i - 1$ and request i , modeled as an exponential random variable with mean 4 hours (here, Y_1 is the time until the first request of the week). For the purposes of this problem, assume that all the $\{X_i\}$ and $\{Y_i\}$ are mutually independent. Let N be the number of requests a customer makes over the course of 7 days (i.e., 168 hours), and let Z be the internal weekly cost to Anthropic for servicing these N requests. Note that N is a discrete random variable (we can't service a fraction of a request).
- (a) (7 points) Determine the conditional expected value $E(Z \mid N)$.
 - (b) (7 points) Determine the expected value of Z . Remember the connection between the exponential and Poisson random variables!
 - (c) (6 points) How much should Anthropic charge for their weekly service to make an average of 100 dollars net profit per month from each customer?